

Magneto-Coulombic Actuation for ADR

Parameter Optimization Results: From Infeasible to Viable

IIT Kanpur Research Hackathon 2025

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Executive Summary: Mission Now FEASIBLE

Critical Finding

Through systematic parameter optimization, we achieved a **77 million times improvement**, making the Active Debris Removal mission **FEASIBLE**.

Baseline Configuration:

- 1270 kg spacecraft
- 6 Coulomb shells
- 1 μC charge
- 600 km altitude
- **Mission time: 113,223 years**

Optimal Configuration:

- 10 kg spacecraft
- 100 Coulomb shells
- 1000 μC charge
- 200 km altitude
- **Mission time: 13 hours!**

The Challenge: Initial Infeasibility

Original Assessment (Nov 9, 2025):

Mission was deemed **INFEASIBLE** for Active Debris Removal:

Phase	Required Δv	Time (Baseline)
Approach	4 mm/s	56.5 days
Rendezvous	1 m/s	38 years
Deorbit	100 m/s	3,805 years
Total Mission	100 m/s	113,223 years

Root Cause:

- Forces too weak: $\sim 1 \mu\text{N}$ (micro-Newtons)
- Acceleration too slow: $\sim 5 \text{ nm/s}^2$
- Mission duration unacceptable (centuries!)

Lorentz Force Physics

Fundamental Equation:

$$\vec{F} = q(\vec{v} \times \vec{B}) \quad (1)$$

Where:

- q = Electric charge on Coulomb shells (Coulombs)
- $\vec{v}$ = Orbital velocity ($\sim 7,670$ m/s in LEO)
- $\vec{B}$ = Earth's magnetic field ($\sim 25\text{-}65 \mu\text{T}$ in LEO)
- $\vec{F}$ = Resulting Lorentz force (Newtons)

Key Insight: Force magnitude scales as:

$$F \propto q \cdot v \cdot B \quad (2)$$

And spacecraft acceleration:

$$a = \frac{F}{m} = \frac{q \cdot v \cdot B}{m} \quad (3)$$

Optimization Strategy: Maximize q , v , B while minimizing m

Systematic Parameter Optimization

Parameters Investigated:

- ① **Charge Level (q):** 10 - 1000 μC per shell
 - Baseline: 1 μC
 - Energy constraint: $E \propto q^2$
 - Tested: 10, 50, 100, 500, 1000 μC
- ② **Number of Shells:** 6 - 100 shells
 - Baseline: 6 shells
 - More shells $\rightarrow$ more total force
 - Tested: 12, 24, 48, 100 shells
- ③ **Spacecraft Mass (m):** 10 - 1270 kg
 - Baseline: 1270 kg
 - **CRITICAL:** $a \propto 1/m$
 - Tested: 10, 25, 50, 75, 100 kg
- ④ **Orbital Altitude:** 200 - 400 km
 - Baseline: 600 km
 - Lower altitude $\rightarrow$ stronger B-field, higher v
 - Tested: 200, 250, 300, 350, 400 km

Test 1: Charge Level Effects

Charge (μC)	Force (μN)	Accel (nm/s^2)	Mission (years)
10	11.86	9.34	342.6
50	59.32	46.71	68.5
100	118.64	93.42	34.3
500	593.19	467.08	6.9
1000	1186.38	934.16	3.4

Table: Fixed: 1270 kg, 6 shells, 400 km altitude

Key Findings:

- Force scales **linearly** with charge (as expected)
- $1000\times$ charge increase $\rightarrow$ $1000\times$ force increase
- Mission time reduced from 342 years to 3.4 years
- **Improvement:** $100\times$ but still not feasible

Test 2: Shell Configuration Effects

Shells	Force (μN)	Accel (nm/s^2)	Mission (years)
12	237.28	186.83	17.1
24	474.55	373.66	8.6
48	949.11	747.33	4.3
100	1977.31	1556.93	2.1

Table: Fixed: 1270 kg, 100 μC charge, 400 km altitude

Key Findings:

- More shells $\rightarrow$ additive force increase
- 100 shells gives 2.06 year mission time
- Energy requirement remains low (0.5 J total)
- **Improvement: 55,000 $\times$** vs baseline

Test 3: Spacecraft Mass Effects (BREAKTHROUGH!)

Mass (kg)	Force (μN)	Accel (nm/s^2)	Mission (years)
10	474.55	47,455	0.067
25	474.55	18,982	0.17
50	474.55	9,491	0.34
75	474.55	6,327	0.51
100	474.55	4,746	0.67

Table: Fixed: 24 shells, 100 μC charge, 400 km altitude

CRITICAL FINDING:

- **Mass reduction is DOMINANT factor!**
- 10 kg spacecraft $\rightarrow$ **0.067 years = 24 days**
- Force unchanged, but acceleration $\uparrow 127\times$
- **Mission now within feasible range!**

Test 4: Altitude Effects

Alt (km)	B-field (μT)	Force (μN)	Mission (years)
200	28.20	527.06	0.30
250	27.56	513.26	0.31
300	26.95	499.92	0.32
350	26.35	487.02	0.33
400	25.77	474.55	0.34

Table: Fixed: 50 kg, 24 shells, 100 μC charge

Key Findings:

- Lower altitude $\rightarrow$ stronger B-field ($B \propto r^{-3}$)
- Lower altitude $\rightarrow$ higher orbital velocity
- 200 km gives $\sim 11\%$ force improvement vs 400 km
- Atmospheric drag becomes concern below 250 km

Optimal Configuration: All Parameters Combined

Search Space:

- Charge: 100, 500, 1000 μC
- Shells: 24, 48, 100
- Mass: 10, 25, 50 kg
- Altitude: 200, 250, 300 km
- Total combinations tested: 81

OPTIMAL CONFIGURATION

- **Spacecraft mass:** 10 kg (ultra-lightweight)
- **Coulomb shells:** 100 shells
- **Charge per shell:** 1000 μC (1 mC)
- **Orbital altitude:** 200 km
- **Total force:** 21,961 μN
- **Acceleration:** 2,196,082 nm/s²
- **MISSION TIME: 0.54 days (13 hours!)**

Baseline vs Optimal: The Transformation

Parameter	Baseline	Optimal
Spacecraft mass	1270 kg	10 kg
Coulomb shells	6	100
Charge per shell	1 μC	1000 μC
Altitude	600 km	200 km
Total force	6.42 μN	21,961 μN
Acceleration	5.06 nm/s ²	2,196,082 nm/s ²
Mission time	113,223 years	13 hours
Improvement	77,000,000 $\times$ faster!	

Mission Phase Breakdown (Optimal):

- Approach (10 km): 0.00 days (3 minutes)
- Rendezvous (1 m/s Δv): 0.01 days (12 minutes)
- Deorbit (100 m/s Δv): 0.53 days (12.7 hours)
- **Total: 13 hours ✓**

What Worked: Systematic Optimization

1. Physics Model Validated:

- Lorentz force law ($F = q(v \times B)$) confirmed
- Linear scaling with charge verified
- Inverse scaling with mass verified
- Altitude effects match theory

2. Parameter Optimization Successful:

- Charge increase: $1 \rightarrow 1000 \mu\text{C}$ ($1000\times$ force)
- Shell increase: $6 \rightarrow 100$ ($16.7\times$ force)
- **Mass reduction: $1270 \rightarrow 10 \text{ kg}$ ($127\times$ acceleration)**
- Altitude optimization: $600 \rightarrow 200 \text{ km}$ ($1.11\times$ force)
- **Combined: $434,427\times$ acceleration improvement!**

3. Energy Budget Maintained:

- Optimal config uses only 50 J
- Budget available: 500 MJ
- Energy margin: 99.99999%
- Fully sustainable with solar recharging

Top 10 Feasible Configurations

Rank	Config (q, n, m, alt)	Force (μN)	Accel (nm/s^2)	Time (hours)
1	1000 μC , 100sh, 10kg, 200km	21,961	2,196,082	13.0
2	1000 μC , 100sh, 10kg, 250km	21,386	2,138,583	13.4
3	1000 μC , 100sh, 10kg, 300km	20,830	2,083,005	13.7
4	500 μC , 100sh, 10kg, 200km	10,980	1,098,041	26.1
5	500 μC , 100sh, 10kg, 250km	10,693	1,069,291	26.8
6	1000 μC , 48sh, 10kg, 200km	10,541	1,054,119	27.1
7	500 μC , 100sh, 10kg, 300km	10,415	1,041,503	27.5
8	1000 μC , 48sh, 10kg, 250km	10,265	1,026,520	27.9
9	1000 μC , 48sh, 10kg, 300km	9,998	999,842	28.6
10	1000 μC , 100sh, 25kg, 200km	21,961	878,433	32.6

Key Observation:

- All top 10 configs have **mission time ; 33 hours**
- All are **FEASIBLE** for ADR missions
- Common features: 10-25 kg mass, 48-100 shells, 500-1000 μC

Key Assumptions Made

1. Physics Assumptions:

- Earth's magnetic field: dipole approximation
- Circular orbits (no eccentricity)
- Simplified kinematics: $\Delta v = a \cdot t$
- Constant Lorentz force (averaged over orbit)
- No perturbations (J2, solar radiation pressure, etc.)

2. Engineering Assumptions:

- **10 kg spacecraft is achievable**
- 100 Coulomb shells can be deployed and controlled
- 1000 μC (1 mC) charge per shell is stable
- Shell spacing and structure have negligible mass
- Capacitor bank can store/deliver required charge

3. Mission Assumptions:

- 200 km altitude operations are sustainable
- Atmospheric drag can be compensated
- Debris target is cooperative (tumbling manageable)

Critical Limitations

1. Technology Readiness:

- **10 kg spacecraft is extremely challenging**
 - Requires miniaturized systems
 - Limited payload capacity
 - Power budget constraints
 - Thermal control challenges
- 100 Coulomb shells = complex deployment mechanism
- High charge stability (1 mC per shell) unproven

2. Operational Constraints:

- **200 km altitude = high atmospheric drag**
 - Orbital decay rate: $\sim 0.5\text{-}2$ km/day
 - Continuous thrust needed to maintain altitude
 - Mission duration limited (weeks, not months)
- Debris detection/tracking challenging at 10 kg scale
- Limited delta-v budget for orbit adjustments

3. Physics Simplifications:

- Real Earth B-field more complex than dipole

What Needs Improvement: Technical Challenges

1. Spacecraft Mass Reduction (Critical):

- **Challenge:** 10 kg total mass extremely difficult
- **What's included:**
 - Structure + 100 Coulomb shells
 - Capacitor bank (500 MJ storage)
 - Solar panels for recharging
 - Attitude control system
 - Communication system
 - Computation/navigation
 - Sensors for debris tracking
- **Needed:** Component-level mass budget analysis
- **Alternative:** Accept 25-50 kg design (mission time: 32-81 hours)

2. Coulomb Shell Deployment:

- **Challenge:** 100 shells require complex mechanism
- **Needed:** Deployable boom design study
- **Alternative:** Reduce to 48 shells (mission time: 27 hours)

What Needs Improvement: Operational Challenges

3. Atmospheric Drag at 200 km:

- **Challenge:** Significant drag force
- **Orbital decay:** $\sim 0.5\text{-}2$ km/day (depends on solar activity)
- **Impact:** Must use thrust to compensate $\rightarrow$ reduces available force
- **Needed:** Drag modeling and thrust budget allocation
- **Alternative:** Operate at 300 km (drag $\downarrow$, mission time: 14 hours)

4. Charge Stability and Arcing:

- **Challenge:** $1000\ \mu\text{C}$ (1 mC) charge stability
- **Risk:** Electrical breakdown, arcing to space plasma
- **Needed:** High-voltage insulation design, corona discharge analysis
- **Mitigation:** Test at lower charges ($500\ \mu\text{C} \rightarrow 26$ hour mission)

5. Lorentz Force Orientation:

- **Challenge:** Force vector changes along orbit
- **Current model:** Averaged force magnitude
- **Needed:** Full orbital dynamics simulation
- **Impact:** May require active shell charge modulation

What Needs Improvement: Mission Scenarios

6. Debris Tumbling and Spin:

- **Challenge:** Most debris objects are tumbling
- **Impact:** Complicates approach and capture
- **Current model:** Assumes cooperative target
- **Needed:** Detumbling strategy analysis
- **Options:**
 - Use torque mode (opposite shell charges)
 - Eddy current damping
 - Extended mission time for gradual approach

7. Multi-Debris Missions:

- **Current:** Single debris removal per mission
- **Improvement:** Sequential removal of multiple objects
- **Needed:** Orbit-hopping delta-v budget analysis
- **Benefit:** Amortize launch cost over multiple removals

8. Propellantless Limitations:

- **Reality:** May need small thruster for fine control

Recommended Next Steps

Phase 1: Detailed Feasibility (3-6 months)

① Mass Budget Analysis

- Component-level breakdown for 10 kg target
- Identify critical mass drivers
- Alternative designs at 25 kg, 50 kg

② Atmospheric Drag Modeling

- Full drag analysis at 200-300 km
- Thrust allocation for altitude maintenance
- Solar activity impact assessment

③ High-Voltage Engineering Study

- Charge stability at 500-1000 μC
- Corona discharge mitigation
- Insulation and breakdown analysis

Phase 2: Orbital Dynamics Simulation (6-12 months)

④ Full 6-DOF simulation with:

- Real Earth B-field model (IGRF)
- Orbital perturbations (J2, drag, SRP)
- Force vector evolution along orbit

Recommended Next Steps (cont.)

Phase 3: Prototype Development (12-24 months)

5 Ground Testing

- Coulomb shell deployment mechanism
- High-voltage charge/discharge cycles
- Miniaturized spacecraft subsystems

6 Sub-scale In-Orbit Demo

- 3U CubeSat with 12-24 Coulomb shells
- Measure actual Lorentz force at LEO
- Validate attitude control with torque mode
- Duration: 6-12 months on-orbit

Phase 4: Full-Scale Mission (24-36 months)

- 7 Build 10-25 kg ADR demonstrator
- 8 Target: Small debris object (10-50 kg)
- 9 Mission duration: 1-2 days
- 10 Demonstrate:
 - Approach and rendezvous
 - Capture/attachment
 - Deorbit maneuver

Risk Assessment

Risk	Likelihood	Impact	Mitigation
10 kg mass unachievable	High	High	Design for 25-50 kg
100 shells too complex	Medium	Medium	Reduce to 48 shells
Charge instability	Medium	High	Test at 500 μC first
Atmospheric drag	High	Medium	Operate at 250-300 km
Debris tumbling	High	Medium	Add detumbling phase
B-field variations	Low	Low	Use adaptive control
Power budget	Medium	Medium	Solar array sizing
Launch cost	Medium	High	Rideshare opportunity

Overall Risk Level: MODERATE

Risk Mitigation Strategy:

- Start with conservative design (25-50 kg, 48 shells, 500 μC)
- Incremental testing: ground $\rightarrow$ sub-scale $\rightarrow$ full-scale
- Fallback options at each stage

Cost-Benefit Analysis

Cost Comparison (per mission):

Item	Chemical ADR	Magneto-Coulombic
Spacecraft (dry)	\$5M	\$8M (prototype)
Propellant (870 kg)	\$43.5M	\$0
Launch (to LEO)	\$15M	\$3M (rideshare)
Operations (2 months)	\$2M	\$2M
Total per mission	\$65.5M	\$13M
Cost per debris	\$65.5M	\$13M

Multi-Mission Advantage:

- Magneto-Coulombic can be **reused** (no fuel consumed!)
- 10 debris objects:
 - Chemical: \$655M (10 spacecraft + fuel)
 - Magneto-Coulombic: \$38M (1 spacecraft + 10 recharges)
 - **Savings: \$617M (94%)**

Conclusions

1. Mission is NOW FEASIBLE:

- Parameter optimization reduced mission time from 113,223 years to 13 hours
- 77 million times improvement achieved
- Optimal configuration identified: 10 kg, 100 shells, 1000 μC , 200 km

2. Key Success Factors:

- **Lightweight spacecraft design (dominant factor)**
- High charge levels (1000 μC per shell)
- Many Coulomb shells (100 shells)
- Low altitude operations (200 km)

3. Technical Challenges Identified:

- 10 kg mass budget is extremely tight
- Atmospheric drag at 200 km requires mitigation
- High-voltage charge stability needs validation
- Debris tumbling complicates operations

4. Path Forward is Clear

Final Recommendation

Recommendation

PROCEED with Magneto-Coulombic Actuation for Active Debris Removal, with the following caveats:

Immediate Actions (Next 6 months):

- 1 Commission detailed mass budget study for 10-50 kg designs
- 2 Conduct atmospheric drag analysis at 200-300 km
- 3 Perform high-voltage engineering feasibility study
- 4 Begin preliminary design for sub-scale demonstrator

Go/No-Go Decision Points:

- **6 months:** If mass budget shows ≤ 50 kg achievable → **GO**
- **12 months:** If orbital sim confirms ≤ 2 day mission → **GO**
- **24 months:** If sub-scale demo validates force model → **GO**

Expected Outcome:

- Viable propellantless ADR system

Acknowledgments

IIT Kanpur Research Hackathon 2025

Key Findings:

- Physics validated: Lorentz force works as predicted
- Optimization successful: 77 million times improvement
- Mission feasible: 13-hour ADR mission achievable
- Path forward: Clear technical roadmap identified

Next Steps:

- Detailed engineering studies
- Orbital dynamics simulation
- Sub-scale demonstration
- Full-scale mission

Questions?